INTRODUCTION TO INFORMATION GEOMETRY

# Gaussian Scores and the Hyperbolic Metric

## Week 10 · Lecture 1

Christopher Lee

# From probability vectors to density curves

A continuous random variable has a density $\rho(x)$ satisfying

$$\rho(x) \geq 0, \qquad \int_{-\infty}^{\infty} \rho(x)\, dx = 1.$$

For an interval $[a, b]$, the probability is its density area:

$$\Pr(a \leq X \leq b) = \int_a^b \rho(x)\, dx.$$

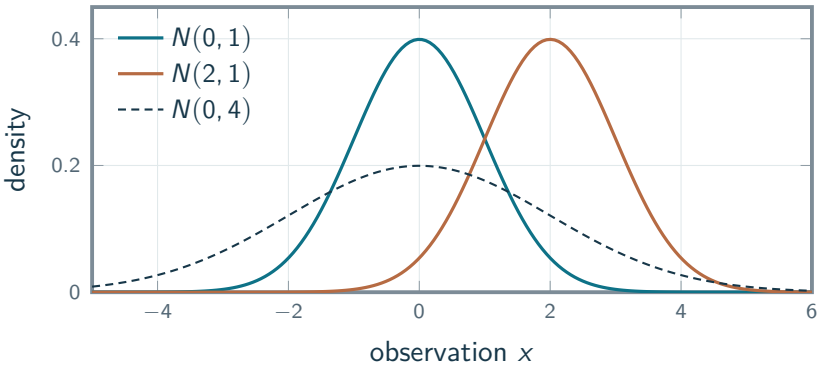
# The two-parameter Gaussian family

The normal density is

$$\rho(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \quad \sigma > 0.$$

The parameter $\mu$ locates the center; $\sigma$ controls the standard deviation. Each point $(\mu, \sigma)$ specifies an entire probability law.

# Moving a point changes a density



# Expectation becomes an integral

For a density $\rho(x)$, the expected value of a function is

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)\rho(x)\,dx.$$

For a Gaussian, introduce the standardized variable

$$Z = \frac{X - \mu}{\sigma}.$$

It has mean zero and variance one.

# The moments we need

Odd standardized moments vanish by symmetry:

$$E[Z] = E[Z^3] = 0.$$

Integration by parts gives

$$E[Z^2] = 1, \qquad E[Z^4] = 3.$$

These four expectations are enough to compute the Gaussian Fisher information matrix.

# The Gaussian log density

Taking the logarithm turns the product and exponential in the Gaussian density into a sum:

$$\ell(x;\mu,\sigma) = -\ln\sigma - \frac{1}{2}\ln(2\pi) - \frac{(x-\mu)^2}{2\sigma^2}.$$

The score in each parameter direction is the derivative of this log density.

# Calculate the two score functions

Starting with

$$\ell = -\ln\sigma - \tfrac{1}{2}\ln(2\pi) - \frac{(x-\mu)^2}{2\sigma^2},$$

find $S_\mu = \partial_\mu \ell$ and $S_\sigma = \partial_\sigma \ell$.

# Calculate the two score functions

Starting with the Gaussian log density,

$$S_\mu = \frac{x-\mu}{\sigma^2}, \qquad S_\sigma = -\frac{1}{\sigma} + \frac{(x-\mu)^2}{\sigma^3}.$$

Equivalently, $S_\mu = Z/\sigma$ and $S_\sigma = (Z^2-1)/\sigma$.

# The scores have mean zero

In standardized form, the two scores are

$$S_\mu = \frac{Z}{\sigma}, \qquad S_\sigma = \frac{Z^2 - 1}{\sigma}.$$

Therefore

$$E[S_\mu] = \frac{E[Z]}{\sigma} = 0, \qquad E[S_\sigma] = \frac{E[Z^2] - 1}{\sigma} = 0.$$

# Compute the Fisher matrix

The Fisher matrix is $G_{ij} = E[S_i S_j]$. Use

$$S_\mu = Z/\sigma, \qquad S_\sigma = (Z^2 - 1)/\sigma$$

and $E[Z^2] = 1$, $E[Z^3] = 0$, $E[Z^4] = 3$. Find the three distinct matrix entries.

# Compute the Fisher matrix

The Fisher matrix is $G_{ij} = E[S_i S_j]$, so

$$G_{\mu\mu} = \frac{1}{\sigma^2}, \qquad G_{\mu\sigma} = 0, \qquad G_{\sigma\sigma} = \frac{3-2+1}{\sigma^2}.$$

Thus

$$G(\mu, \sigma) = \frac{1}{\sigma^2} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

# The Gaussian Fisher line element

For a small parameter change $(d\mu, d\sigma)$,

$$ds_F^2 = \frac{d\mu^2 + 2\,d\sigma^2}{\sigma^2}.$$

Mean and scale directions are orthogonal in this metric. The factor $1/\sigma^2$ makes a fixed numerical change statistically longer when the distribution is narrow.

# Compare two small Gaussian changes

At $N(0,4)$, the standard deviation is $\sigma = 2$. Compare the Fisher lengths of

$$(d\mu, d\sigma) = (h, 0) \quad \text{and} \quad (0, h).$$

Which change has greater length for the same $|h|$?

# Compare two small Gaussian changes

At $\sigma = 2$, the squared lengths are

$$\frac{h^2}{4} \quad \text{and} \quad \frac{2h^2}{4}.$$

Thus their lengths are

$$\frac{|h|}{2} \quad \text{and} \quad \frac{|h|}{\sqrt{2}}.$$

A scale change is $\sqrt{2}$ times as long.

# Rescale the horizontal coordinate

Set

$$u = \frac{\mu}{\sqrt{2}}, \qquad v = \sigma > 0.$$

The line element becomes

$$ds_F^2 = 2\frac{du^2 + dv^2}{v^2}.$$

The coordinate region is the upper half-plane $v > 0$.

# A hyperbolic geometry emerges

The Poincaré half-plane metric is

$$ds^2_{\mathbb{H}} = \frac{du^2 + dv^2}{v^2}.$$

The Gaussian Fisher metric is twice this metric. Lengths scale by $\sqrt{2}$; its constant Gaussian curvature is $K = -1/2$.

# Why traveling higher can be shorter

The length of a parameter path $(u(t), v(t))$ is

$$L_F = \sqrt{2} \int \frac{\sqrt{u'(t)^2 + v'(t)^2}}{v(t)}\, dt.$$

At larger $v$, the same Euclidean displacement contributes less Fisher length. Geodesics use that variation in cost to connect distributions.

# Two spaces, two curvatures

The interior of a finite categorical family is mapped to part of a sphere of radius 2. Its Fisher curvature is $K = +1/4$. The full Gaussian family becomes a hyperbolic half-plane with $K = -1/2$. Both metrics come from the same construction: expected products of score functions.